

2011 Year 12
Semester 1 Examination

Mathematics Extension 1

General Instructions

- All questions should be attempted
- Start each 'Question' on a new page
- Answers without appropriate working and/or diagrams may not attract full marks
- Approved silent calculators may be used

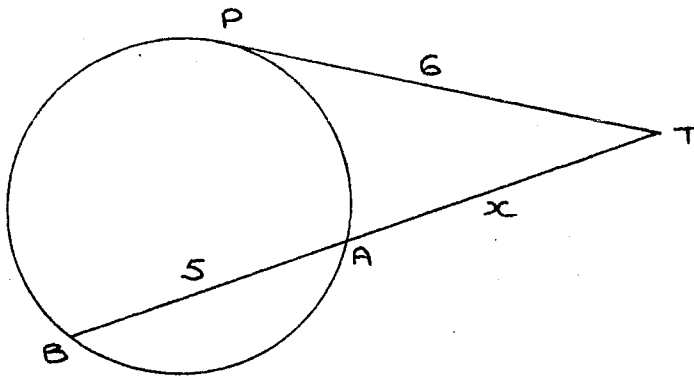
Time Allowed

1½ hours + 5 minutes reading time

Question 1 (13 marks)

MARKS

- a) Convert an angle of 1.85 radians to degrees. 1
- b) If α, β, γ are the roots of the equation: $x^3 - 5x^2 + 2x - 1 = 0$
find the value of:
- (i) $\alpha + \beta + \gamma$ 1
- (ii) $\alpha \beta \gamma$ 1
- (iii) $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$ 1
- c) 2



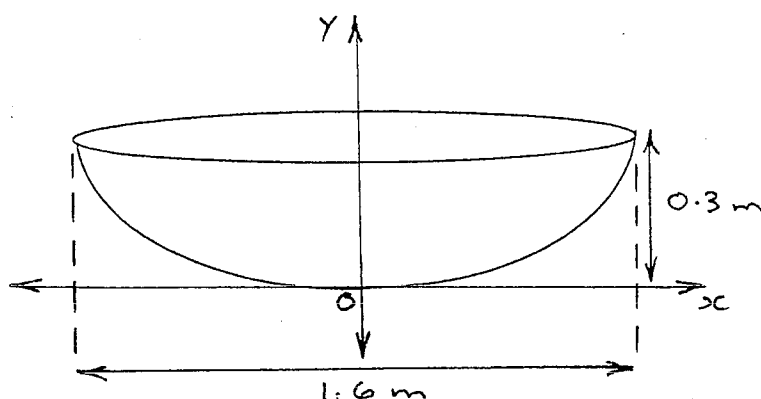
PT is a tangent and TAB is a secant.
Find the value of x .

- d) Find $\int x \cdot e^{-x^2} dx$

1

- e) A satellite dish in the shape of a paraboloid has its vertex at the origin. Its diameter is 1.6 metres and its depth 0.3 metre as shown.

3



Find the equation of the parabola required, in the form $x^2 = 4ay$. Hence find the satellite dish's focal length 'a'.

f) Find $\int \frac{1+5x^2-x^3}{x^2} dx$

2

g) What is the exact value of $\sec \frac{\pi}{6}$?

1

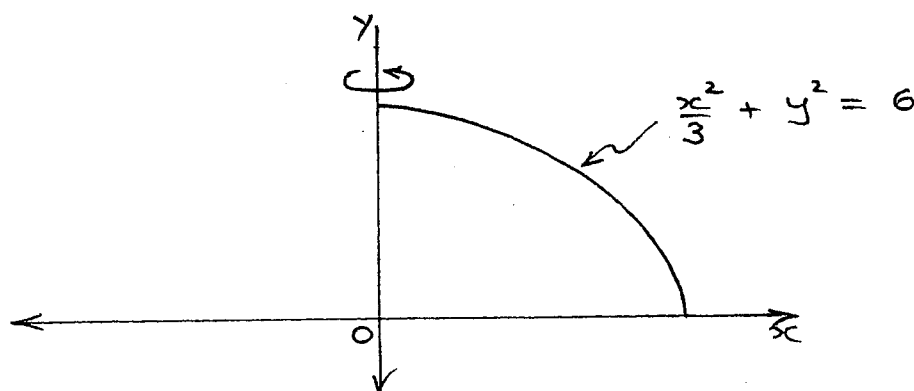
Question 2 (13 marks)

a) Perform the long division: $(x^3 - 2) \div (x + 1)$
writing your answer in the form $A(x) + \frac{R}{x+1}$

2

b)

4

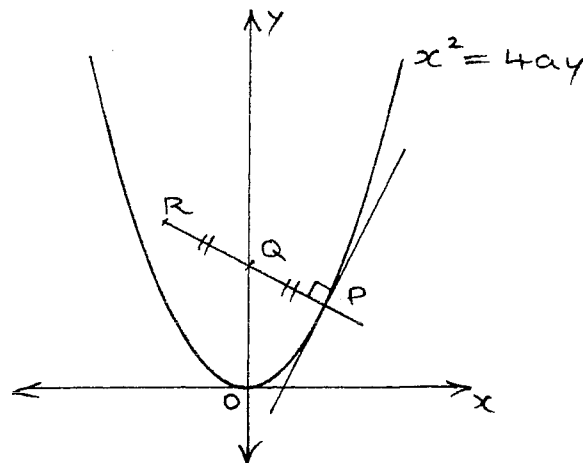


That part of the curve $\frac{x^2}{3} + y^2 = 6$ that lies in the first quadrant is rotated about the y axis. Find the exact volume of the solid of revolution.

c) Given that $y = e^{kx}$ is a solution to $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = 0$
find the values of k .

3

- d) The normal at the point $P(2at, at^2)$ on $x^2 = 4ay$ cuts the y axis at Q , and is so produced to R such that $PQ = QR$.



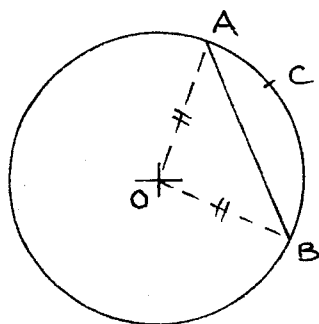
- (i) You are given that the normal at P has the equation $x + ty - at(t^2 + 2) = 0$ (you are not required to prove this). Find the co-ordinates of R in terms of t . 2
- (ii) Find the cartesian locus of R . 2

Question 3 (13 marks)

- a) A function $y = f(x)$ had just one stationary point, at $x = 3$. Using the table of results shown below, what is the nature of the stationary point? 1

x	2	3	4
$f'(x)$	-0.1	0	-5

- b)



AB is a chord of length 10cm, in a circle of radius 6cm.

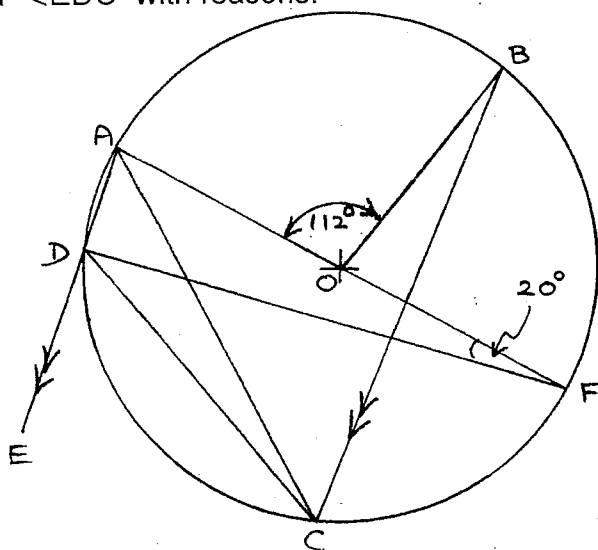
- (i) Find $\angle AOB$ in radians (correct to 3 significant figures). 2
- (ii) Find the perimeter of segment ABC. 1
- (iii) Find the area of segment ABC 1

- c) Solve the equation: $x^3 - 3x^2 - 4x + 12 = 0$ given that the sum of two of its roots is zero. 2
- d) P is the point $(2at, at^2)$ on the parabola $x^2 = 4ay$. S is the focus. PR is drawn parallel to the parabola's axis, meeting the directrix in R. Prove that RS is parallel to the normal at P. 3
- e) Starting with an initial value of $x = 1$, use one application of Newton's Method to find a better approximation to the root of the equation: $2x - \ln(x + 3) = 0$. 3

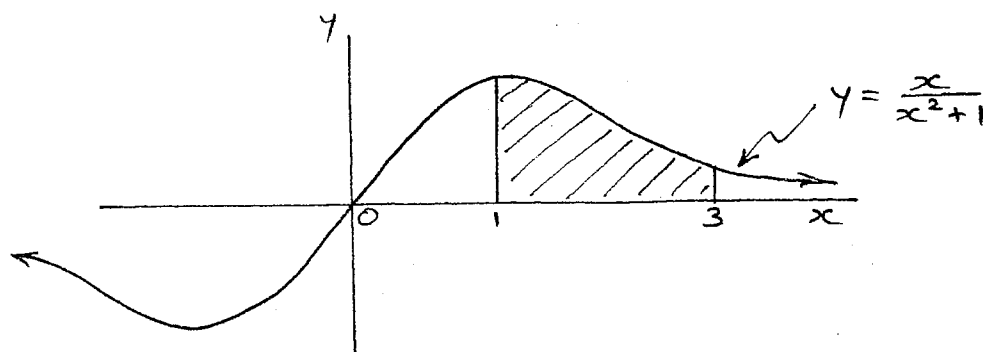
Question 4 (13 marks)

- a) i) Copy the diagram below into your examination booklet.

- ii) O is the centre of the circle and AF is a diameter. Find $\angle EDC$ with reasons. 3



- b) Solve for x correct to three significant figures: $2^x = 40$ 2
- c) Find $\int \frac{dx}{\sqrt[3]{2x+1}}$ 2
- d) Solve for x : $2 \ln x = \ln(x + 6)$ 3
- e) Calculate the shaded area shown. 3



Question 5 (13 marks)

For the curve with equation $y = \frac{x+1}{(x-1)^2}$

- (i) What is its natural domain? 1
- (ii) Show that $\frac{dy}{dx} = \frac{-x-3}{(x-1)^3}$ 2
- (iii) Find the one stationary point this curve has, and determine its nature. 3
- (iv) You are given its second derivative as 2
- $$\frac{d^2y}{dx^2} = \frac{2x+10}{(x-1)^4}$$
- Prove that there is a point of inflexion at $x = -5$.
- (v) What happens to the value of y as x approaches infinity? 1
- (vi) Draw a neat half page sketch of the curve showing all relevant information. 3
- (vii) Determine the range of the above function. 1

----- END OF PAPER -----

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1 a) 106°

b) i) $\frac{-(-5)}{(1)} = 5$

ii) $\frac{-(-1)}{(1)} = 1$

iii) $\frac{1}{\alpha\beta} + \frac{1}{\alpha\delta} + \frac{1}{\beta\delta}$
 $= \frac{\delta + \beta + \alpha}{\alpha\beta\delta}$

$= \frac{5}{1}$
 $= 5$

c) $x(x+5) = 6^2$

$x^2 + 5x - 36 = 0$

$(x+9)(x-4) = 0$

$x = -9, 4$

$\therefore x = 4$ only since $x > 0$

d) $I = -\frac{1}{2} \int -2x \cdot e^{-x^2} dx$
 $= -\frac{1}{2} e^{-x^2} + C$

e) Curve passes thru' $(0.8, 0.3)$

$\therefore (0.8)^2 = 4a(0.3)$

$0.64 = 1.2a$

$a = 0.5\dot{3}$ is focal length

and the curve's equation is

$x^2 = 2.1\dot{3}y$

f) $I = \int (x^{-2} + 5 - x) dx$
 $= \frac{x^{-1}}{-1} + 5x - \frac{x^2}{2} + C$

$= -\frac{1}{x} + 5x - \frac{x^2}{2} + C$

g) $\frac{2}{\sqrt{3}}$ (or $\frac{2\sqrt{3}}{3}$)

Q2 a)

$$\begin{array}{r} x^2 - x + 1 \\ x+1 \overline{) x^3 - 2} \\ \underline{x^3 + x^2} \\ -x^2 - 2 \\ \underline{-x^2 - x} \\ x - 2 \\ \underline{x + 1} \\ -3 \end{array}$$

$\therefore \frac{x^3 - 2}{x+1} = x^2 - x + 1 - \frac{3}{x+1}$

b) $x^2 = 18 - 3y^2$

Also, when $x=0$, $y = \sqrt{6}$

$\therefore V = \pi \int_0^{\sqrt{6}} (18 - 3y^2) dy$

$= \pi [18y - y^3]_0^{\sqrt{6}}$

$= \pi \{ [18\sqrt{6} - 6\sqrt{6}] - [0] \}$

$= 12\pi\sqrt{6} \text{ (u3)}$

c) $y = e^{kx}$ $y' = ke^{kx}$ $y'' = k^2 e^{kx}$

$\therefore k^2 e^{kx} - 4ke^{kx} + 3e^{kx} = 0$

$e^{kx} (k^2 - 4k + 3) = 0$

$e^{kx} \neq 0 \therefore (k-3)(k-1) = 0$

$k = 1, 3$

d) i) let $x=0$

$\therefore ty - at(t^2+2) = 0$

$y = a(t^2+2)$

$\therefore Q$ is $(0, at^2+2a)$

Let R be (x, y)

$\therefore \frac{x+2at}{2} = 0 \rightarrow x = -2at$

$y + at^2 = at^2 + 2a$

$y + at^2 = 2at^2 + 4a$

$y = at^2 + 4a$

$\therefore R$ is $(-2at, at^2 + 4a)$

ii) $\begin{cases} x = -2at & \text{--- ①} \\ y = at^2 + 4a & \text{--- ②} \end{cases}$

From ① $t = \frac{-x}{2a}$

Sub. into ② $y = a\left(\frac{-x}{2a}\right)^2 + 4a$

i.e. $y = \frac{x^2}{4a} + 4a$ is the

cartesian locus of R.

Q3. a) $\searrow \rightarrow \swarrow$

Hence a stationary inflexion is present at $x=3$.

$$\begin{aligned} \text{b) i) } \cos \theta &= \frac{6^2 + 6^2 - 10^2}{2 \times 6 \times 6} \\ &= \frac{-7}{18} \end{aligned}$$

$$\begin{aligned} \theta &= \cos^{-1}\left(\frac{-7}{18}\right) \text{ (in radian mode)} \\ &\doteq 1.97 \text{ (3sf)} \end{aligned}$$

$$\begin{aligned} \text{ii) } P &= 6 \times 1.97 + 10 \\ &= 21.8 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{iii) } A &= \frac{1}{2} \times 6^2 (1.97 - \sin 1.97) \\ &\doteq 18.875 \text{ cm}^2 \end{aligned}$$

$$\text{c) } \alpha + \beta + \delta = \frac{-(-3)}{(1)}$$

$$\begin{aligned} \therefore 0 + \delta &= 3 \\ \delta &= 3 \end{aligned}$$

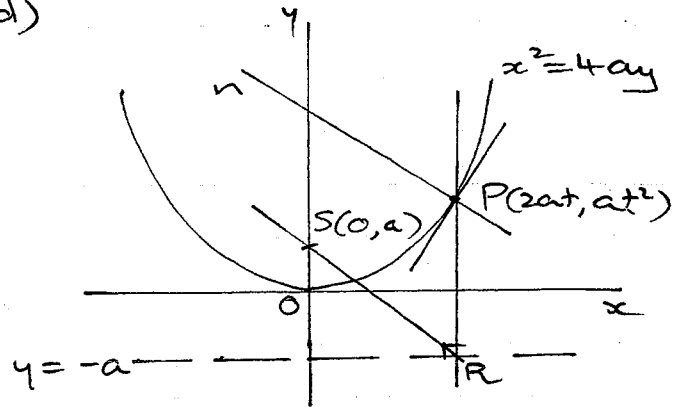
$$\text{Also, } \alpha\beta\delta = \frac{-(12)}{(1)}$$

$$3\alpha\beta = -12$$

$$\alpha\beta = -4$$

$$\therefore \alpha = 2, \beta = -2, \delta = 3$$

d)



$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{1}{\frac{dx}{dt}} = 2at \times \frac{1}{2a} = t$$

$$\therefore m_n = -\frac{1}{t}$$

R has coord's $(2at, -a)$

$$\therefore m_{RS} = \frac{a - -a}{0 - 2at}$$

$$= \frac{2a}{-2at}$$

$$= -\frac{1}{t}$$

$$= m_n$$

$\therefore RS \parallel$ normal at P.

e) Let $f(x) = 2x - \ln(x+3)$

$$\therefore f'(x) = 2 - \frac{1}{x+3}$$

$$a_1 = 1 - \frac{[2(1) - \ln(1+3)]}{[2 - \frac{1}{1+3}]}$$

$$= 1 - \frac{2 - \ln 4}{1\frac{3}{4}}$$

$$\doteq 0.649$$

Q4

a) i) —

ii) $\angle ACD = 20^\circ$

($\angle$'s on common arc AD)

$$\begin{aligned} \angle ACB &= \frac{1}{2} \times 112^\circ \\ &= 56^\circ \end{aligned}$$

($\angle$ at circ. $\frac{1}{2} \times \angle$ at centre, on common arc AB)

$$\angle DCB = 20^\circ + 56^\circ$$

$$= 76^\circ$$

(addition of adj. $\angle$'s)

$$\therefore \angle EDC = 76^\circ$$

(alt. $\angle$'s in $\parallel$ lines)

$$b) \quad x = \log_2 40$$

$$= \frac{\ln 40}{\ln 2}$$

$$= 5.32192 \dots$$

$$\approx 5.32$$

$$c) \quad I = \int (2x+1)^{-\frac{1}{3}} dx$$

$$= \frac{(2x+1)^{\frac{2}{3}}}{\frac{2}{3}} + C$$

$$= \frac{3}{4} \cdot \sqrt[3]{(2x+1)^2} + C$$

$$d) \quad \ln x^2 = \ln(x+6)$$

$$\therefore x^2 = x+6$$

$$x^2 - x - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$\therefore x = 3 \text{ only}$$

($x > 0$ in the expression $2 \ln x$)

$$e) \quad A = \int_1^3 \frac{x}{x^2+1} dx$$

$$= \frac{1}{2} \int_1^3 \frac{2x}{x^2+1} dx$$

$$= \left[\frac{1}{2} \ln(x^2+1) \right]_1^3$$

$$= \frac{1}{2} [\ln 10 - \ln 2]$$

$$= \frac{1}{2} \ln \left(\frac{10}{2} \right)$$

$$= \frac{1}{2} \ln 5 \quad (u^2).$$

$$Q5. \quad y = \frac{x+1}{(x-1)^2}$$

$$i) \quad D = \{ \text{all } x \neq 1 \}$$

$$ii) \quad u = x+1 \quad v = (x-1)^2$$

$$u' = 1 \quad v' = 2(x-1) \cdot 1$$

$$= 2(x-1)$$

$$y' = \frac{(x-1)^2 \cdot 1 - (x+1) \cdot 2(x-1)}{[(x-1)^2]^2}$$

$$= \frac{(x-1)[(x-1) - 2(x+1)]}{(x-1)^4}$$

$$= \frac{x-1-2x-2}{(x-1)^3}$$

$$= \frac{-x-3}{(x-1)^3} \quad \text{as req'd}$$

$$iii) \quad \text{let } y' = 0$$

$$\therefore -x-3 = 0$$

$$x = -3$$

$$\rightarrow y = \frac{(-3)+1}{((-3)-1)^2} = -\frac{1}{8}$$

$$x \quad -4 \quad -3 \quad -2$$

$$y' \quad - \quad 0 \quad +$$

$\therefore (-3, -\frac{1}{8})$ is a local min.

$$iv) \quad \text{let } y'' = 0 \text{ for a possible inflex.}$$

$$\therefore 2x+10 = 0$$

$$x = -5$$

$$x \quad -6 \quad -5 \quad -4$$

$$y'' \quad -0.0008 \quad 0 \quad 0.0032$$

$\therefore$ there is a change in concavity

$\therefore$ an inflexion exists at $x = -5$

$$(\text{also, } y = \frac{-4}{(-6)^2} = -\frac{1}{9})$$

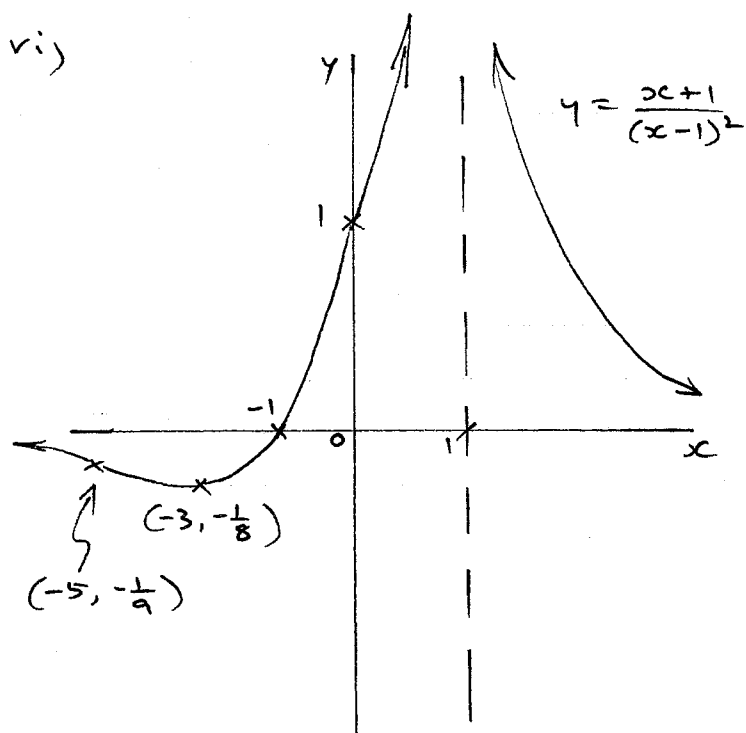
v) As $x \rightarrow \infty$, $y \rightarrow 0$

(note that $\frac{x+1}{(x-1)^2}$ becomes

$$\frac{x+1}{x^2-2x+1} = \frac{\frac{1}{x} + \frac{1}{x^2}}{1 - \frac{2}{x} + \frac{1}{x^2}} = \frac{0}{1} = 0$$

as $x \rightarrow \infty$)

vi)



vii) $R = \{y \geq -\frac{1}{8}\}$